

Three passive monitor-only runs investigating the origin of the repeatable ~33-min thermal/spectral oscillation at the EXOhSPEC camera. **V19** (5 h) established the LK220-mediated cooling-chain hypothesis. **V20** (2 h) added a fourth PT104 probe directly on the camera rear (between the two coolant connections) with fully-confirmed LK220 telemetry. **V21** (2 h) repeated V20 with the camera's own CMOS TEC switched OFF, LK220 left ON, to isolate the camera's cooling duty cycle as a load modifier versus the oscillation's true origin. All statistics below are recomputed independently from the raw CSVs.

Bottom line

Culprit: the LK220, reacting to the room AC — not the camera. Chain is AC/room cycle → LK220's own control loop → coolant → camera body → pixel position. **Proof:** in V21 the camera TEC was fully OFF (0% cooling demand, doing nothing) for the whole run, and the LK220's own current/coolant temperature kept oscillating at the same ~32-min period anyway — the camera cannot be driving something happening while it is switched off.

Camera TEC on vs off: OFF genuinely shrinks the local heat excess at the camera body (left probe +1.85°C→+0.17°C above ambient) and lowers the LK220's total load — real wins. But it does **not** quiet the oscillation itself (residual noise 0.203→0.224 px, slightly worse), while the mean pixel position shifts far more from the (0,0) reference (0.582→1.110 px, most likely just the sensor sitting 34°C warmer, not a noise effect). **Net verdict: keep the camera TEC on** — the fix has to happen upstream at the LK220/AC, not by touching the camera.

V19 — 5-hour passive baseline

Aim. Investigate why a strong AC-related oscillation appears in the cooling/camera temperatures while the internal instrument environment stays comparatively stable. **Dataset.** 301 frames over 5.00 h; 3612 LK220 samples at ~5 s cadence; bench TEC fixed at 22°C, no AO correction.

Bench TEC 22.001±0.012°C. Internal ECU 23.586±0.008°C (0.034°C p-p — very stable). BME 23.623±0.145°C (includes initial settling). PT104 AC 22.642±0.141°C, 0.535°C p-p, clearly periodic. Camera left/right/top ran **+1.34, +0.86, +1.72°C** above ECU (**+2.29, +1.81, +2.67°C** above the AC probe) — the camera is locally hotter than its surroundings and its oscillation is far larger than what the internal ECU records, so uniform box-air heating cannot explain it alone.

Source/channel	Mean	σ	P-P	r(L)	r(R)	r(Top)
Bench TEC	22.001°C	0.012	0.073	—	—	—
Internal ECU	23.586°C	0.008	0.034	0.104	0.154	0.119
BME/HVAC T	23.623°C	0.145	1.370*	0.089	0.067	0.143
PT104 AC probe	22.642°C	0.141	0.535	0.345	0.449	0.459
Camera left	24.928°C	0.104	0.396	—	—	—
Camera right	24.450°C	0.075	0.314	—	—	—
Camera top	25.310°C	0.133	0.530	—	—	—
LK220 TACT	21.993°C	0.149	0.490	0.896	0.902	0.907
LK220 TEC current	4.00 A	0.93	3.30	0.367	0.474	0.504
CoolerPower	72.58%	0.55	2.00	0.873	0.842	0.854

*BME peak-to-peak includes initial settling. dX_RMS = 0.076 px, dY_RMS = 0.537 px, dY within ±0.5 px = 54.2%, dY within ±0.3 px = 34.2%, OPL vs dY r = -0.45. Common period ≈ 33.4 min across AC, LK220 TACT/current, camera temps, CoolerPower, OPL and dY.

Working interpretation

AC oscillation → LK220 thermal response → coolant/camera heat sink → camera cooler response → camera-body temperature → optical/spectral drift, rather than uniform box-air heating.

Next step that motivated V20/V21

Place a PT104 probe directly on the camera heat sink beside the coolant connections, to test whether the sequence becomes LK220 TACT → heat sink → camera L/R/top → OPL/dY.

V19 — diagnostic plots

V19: full-run diagnostic (PT104 channels merged into one thermal panel)

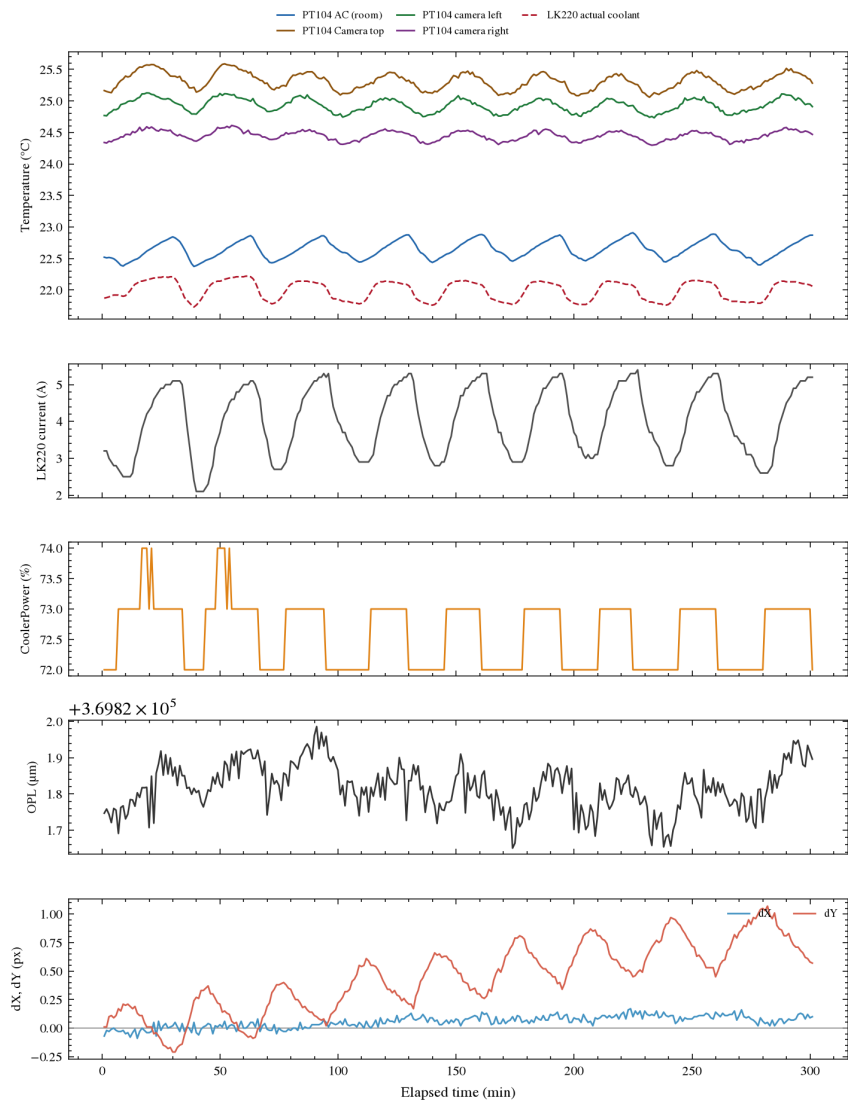


Figure V19.1. Full-run diagnostic, PT104 AC/top/left/right and LK220 actual coolant merged into one thermal panel (was 5 separate rows). The repeated cycle is visible together there, and in LK220 current, CoolerPower and dY, while the internal ECU (not shown here, see Fig. V19.4) stays comparatively stable.

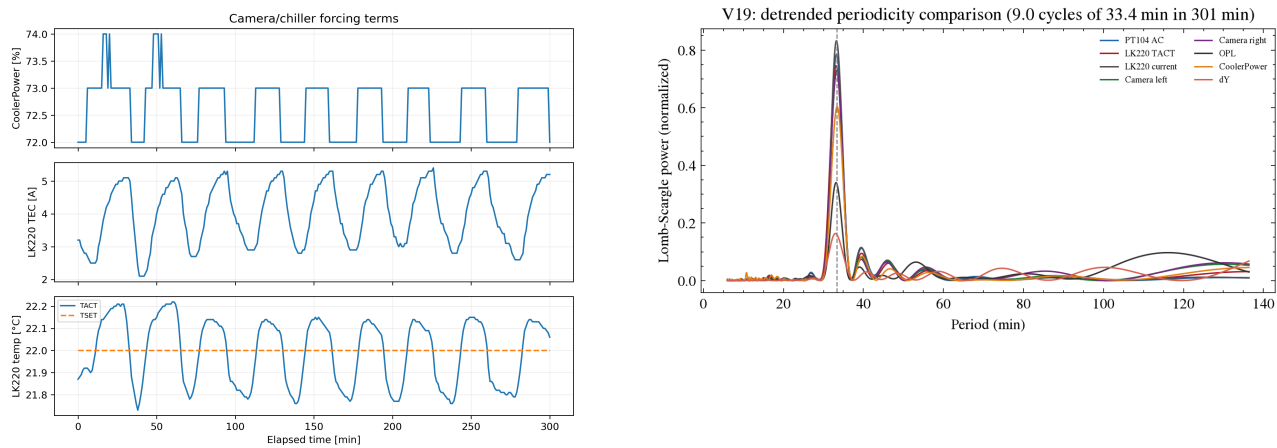


Figure V19.2 (left). CoolerPower, LK220 TEC current, LK220 TACT. **Figure V19.3 (right).** Detrended periodicity comparison, dominant common component near 33.4 min.

V20 — 2-hour repeat, confirmed LK220 telemetry, rear probe added

Aim. Repeat V19 conditions (camera TEC ON at −10°C) with fully-parsed LK220 MONITOR? telemetry and PT104 Ch4 relocated to the **camera rear, between the two coolant connections** (Bill's suggestion), to directly test the missing coolant–camera heat-sink link. **Dataset.** 121 frames over 2.00 h; 1452 LK220 samples at ~5 s.

Bench TEC 21.9997±0.0118°C. ECU 23.606±0.025°C. PT104 AC 22.612±0.152°C (0.612°C p-p). Camera left 25.455±0.135°C, right 24.392±0.080°C, **rear (new) 23.115±0.131°C**. Rear sits **−0.49°C** relative to ECU (cooler than ambient — it sits right at the coolant lines) while left/right remain **+1.85/+0.79°C** hotter: a clean spatial gradient coolant → heat sink → camera body. LK220 TACT 22.044±0.139°C (0.390°C p-p), TEC current 3.978±0.930 A (2.3–5.2 A), TEC duty 31.82±5.25% (new this run). CoolerPower 72.64±0.56% — essentially identical to V19's 72.58%, a good night-to-night consistency check.

Source/channel	Mean	σ	P-P	r(L)	r(R)	r(Rear)
Bench TEC	21.9997°C	0.0118	0.055	–	–	–
PT104 AC probe	22.612°C	0.152	0.612	−0.021	0.081	−0.116
BME/HVAC T	22.926°C	0.146	1.100*	−0.170	−0.181	−0.166
Camera left	25.455°C	0.135	0.481	–	–	–
Camera right	24.392°C	0.080	0.344	–	–	–
Camera rear (new)	23.115°C	0.131	0.484	–	–	–
LK220 TACT	22.044°C	0.139	0.390	0.667	0.690	0.611
LK220 TEC duty (new)	31.82%	5.25	16.2	see text	see text	see text
CoolerPower	72.64%	0.56	2.00	0.844	0.806	0.814

*BME p-p includes settling, as in V19. dX_RMS = 0.039 px, dY_RMS = 0.568 px, dY within ±0.5 px = 49.6%, within ±0.3 px = 18.2%, OPL vs dY r = 0.664. AC-vs-camera correlations flipped sign relative to V19 (there +0.35–0.46); with only ~2.6–3.1 cycles of the 33.4-min oscillation in 2 h, zero-lag Pearson r on a quasi-periodic signal is highly sensitive to the window's phase — read as a short-duration sampling artefact, not a changed physical link. CoolerPower and LK220 TACT stay the robust predictors in both runs.

Lag structure (±15 min window, to avoid locking onto an adjacent 33-min cycle): AC leads LK220 TACT by ~3 min (r=0.82). Rear leads camera-left by only ~1 min (r=0.99); camera-left leads CoolerPower by ~1 min (r=0.85) — both effectively simultaneous. The LK220–rear step itself was ambiguous at large lag; only the zero-lag value (r=0.61) is reported for that step.

Main result

The rear probe closes the V19 gap: it sits within 0.5°C of the coolant and leads camera-left by ~1 min (r=0.99). CoolerPower remains the single strongest, most reproducible correlate of camera-body temperature across V19 and V20 (r=0.81–0.87).

Open question carried into V21

V20 alone cannot separate “TEC duty causes the heat excess” from “TEC duty merely correlates with it because both respond to the same LK220/AC cycle.” V21 switches the camera TEC off to test this directly.

V20 — diagnostic plots

V20: full-run diagnostic (PT104 channels merged into one thermal panel)

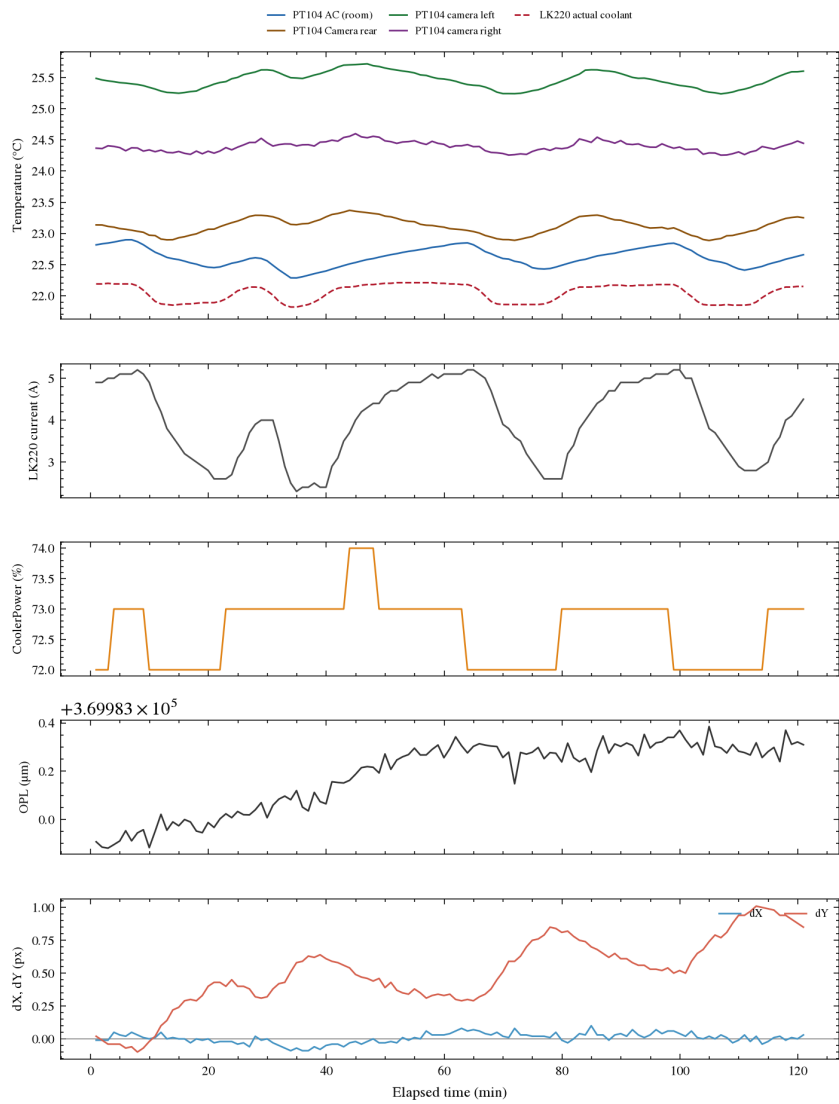


Figure V20.1. Full-run diagnostic, all 4 PT104 channels + LK220 actual coolant merged into one thermal panel; bottom panel shows dX and dY together.

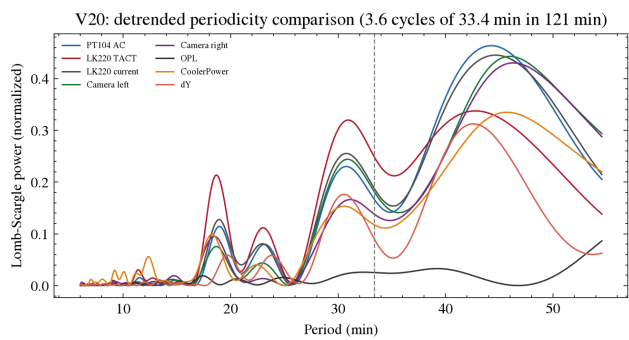
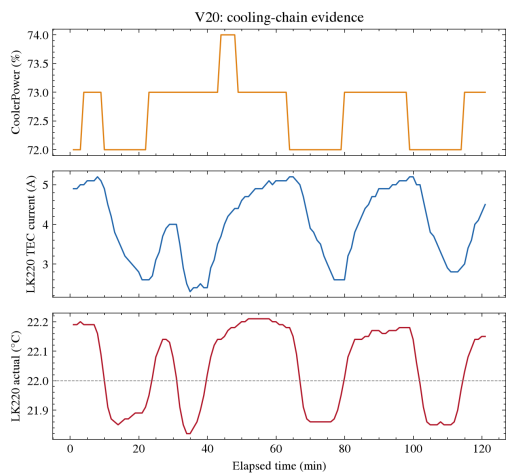


Figure V20.2 (left). CoolerPower, LK220 current, LK220 TACT. Figure V20.3 (right). Periodicity comparison (only ~2.6–3.1 cycles fit in this 2 h run, so treat the exact peak as indicative).

V21 — camera TEC OFF, LK220 still ON (controlled repeat of V20)

Aim. Repeat V20 with the camera's CMOS TEC switched **OFF**, LK220 left ON, to isolate whether the camera's own cooling duty is the *origin* of the oscillation or a downstream responder. **Dataset.** 121 frames, 2.00 h; CoolerOn=False, CoolerPower=0% for the entire run.

Thermal-step, not a steady baseline — frame 0 was taken right after the TEC-off switch. Fit $T(t)=T_{eq}-A\cdot e^{-t/\tau}$ to CMOS: **$T_{eq}=24.02\pm0.02^{\circ}\text{C}$, $A=17.12\pm0.35^{\circ}\text{C}$, $\tau=2.51\pm0.06\text{ min}$** . All steady-state numbers below use $t\geq 20\text{ min}$ ($\approx 8\tau$), applied identically to V20 for a fair comparison.

Post-20-min: camera left **+0.17°C** vs. ECU (was +1.85°C in V20), right **-0.21°C** (was +0.79°C), rear **-1.04°C** (was -0.49°C) — the local camera heat excess largely disappears with the TEC off.

Quantity	V20 (TEC ON)	V21 (TEC OFF)
CMOS temp	-9.99±0.04°C	24.02±0.02°C
CoolerPower	72.6±0.6%	0%
LK220 heat sink	31.39±1.84°C	29.40±1.74°C
LK220 TEC current	3.93±0.93 A	2.67±0.94 A
LK220 TEC duty	31.5±5.2%	24.3±5.3%
LK220 flow (confound check)	0.757 LPM	0.756 LPM
Camera rear	23.13±0.13°C	22.26±0.14°C
dY mean offset	0.582 px	1.110 px
dY std about own mean	0.203 px	0.224 px

Flow unchanged — rules out a flow-rate confound. Heat sink/current/duty all measurably lower with TEC off: the LK220's total load genuinely dropped. But dY's residual scatter about its own mean barely moves (0.203→0.224 px, ~10%) even though the mean offset nearly doubled (most plausibly a CMOS-package mechanical offset from running 34°C warmer, not a noise effect). **Turning the camera TEC off did not meaningfully quiet the oscillation.**

Most direct evidence in this dataset

With CoolerPower pinned at exactly 0% for the whole run, LK220's own TEC current still swings 1.0–4.9 A and TACT still oscillates ±0.2°C on the same ~32-min period. The camera cannot be driving this — it is doing nothing. The oscillation's origin is upstream, in the LK220/room-AC loop, now demonstrated under a condition where the camera is provably not the source.

Updated interpretation & caveats

AC/room → LK220 (self-sustaining, camera-independent) → coolant → camera body → OPL/dY. Camera-TEC duty scales the *magnitude* of local heat at the mount; the ~33-min *clock* lives upstream.

Caveats: V20/V21 ran on different nights (ECU 23.61 vs 23.32°C) — not a perfectly isolated single-variable swap. Several lags hit the ±15 min search boundary and are excluded as ambiguous. Correlation and shared periods do not by themselves prove causality.

V21 — diagnostic plots

V21: full-run diagnostic (PT104 channels merged into one thermal panel)

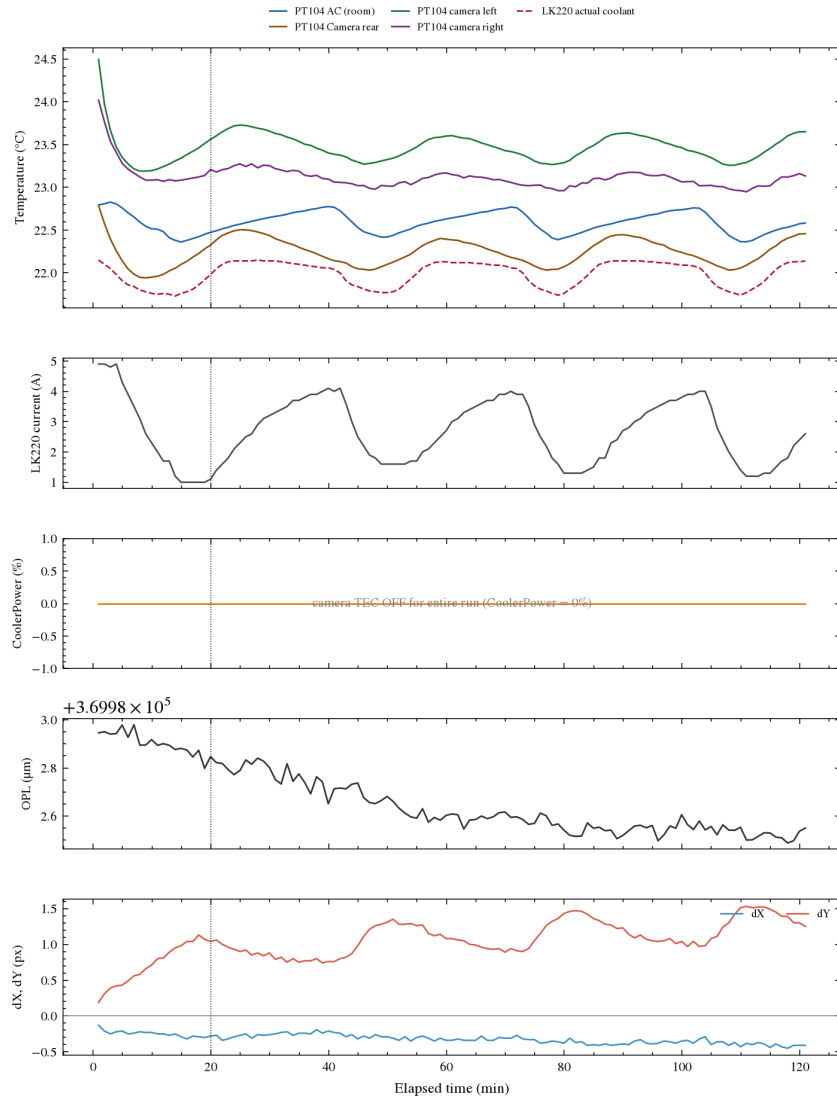


Figure V21.1. Dotted line at t=20 min marks the defensible steady-state start from the warm-up fit. CoolerPower panel is flat at 0% — not a plotting artefact.

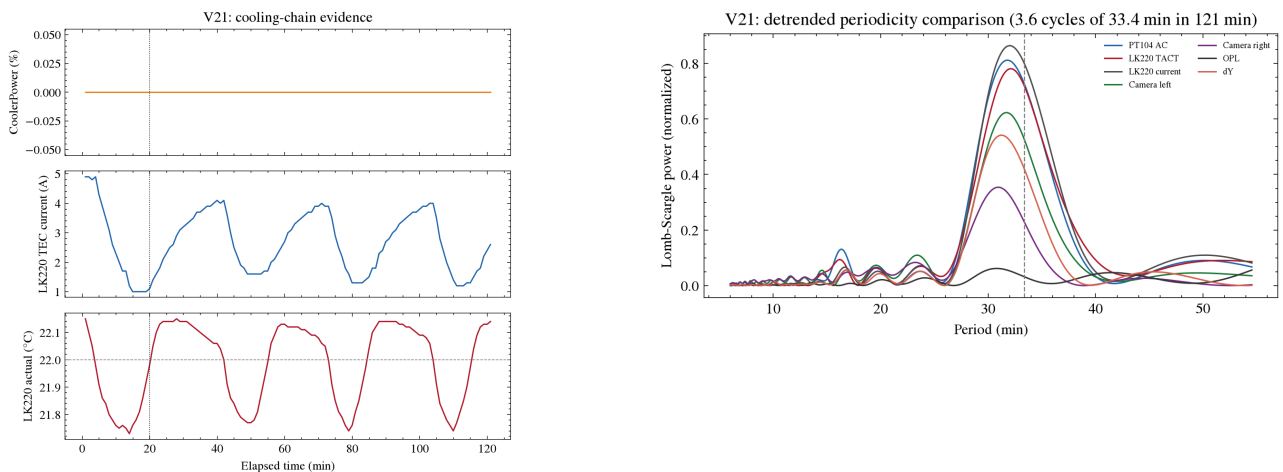


Figure V21.2 (left). CoolerPower flat at zero while LK220 current/TACT keep oscillating — direct evidence the oscillation originates upstream of the camera.
Figure V21.3 (right). Periodicity comparison, CoolerPower omitted (constant zero has no spectrum); ~3.6–3.9 cycles span this 2 h run.

Next decisive measurement

V21 shows the LK220/room loop drives the oscillation independent of the camera. The decisive follow-up moves one level further upstream: attach the LK220's external TSP-TH sensor near the AC intake / room air (SENS=1, external control sensor per the LK220 manual) so the chiller regulates against the load point rather than its own internal return-line thermistor, and repeat. In parallel, run a full 5-hour TEC-off repeat matching V19's duration, so the ~33-min period is resolved over ~9 cycles instead of the current ~3.6–3.9.